

1. Resolve Merge Conflicts

- Create a merge conflict intentionally (two users editing the same line).
- Resolve the conflict and push the changes.

- ➔ Created a file in Ravutla branch with some content
- ➔ Committed the file and pushed it to the repository
- ➔ Created another file in Poojith branch with same name and content.
- ➔ Committed the file and pushed in to repository.
- ➔ Tried to merge both the branches
- ➔ It Raised a conflict
- ➔ Checked that which file is raising the conflict using “git status”
- ➔ Modified the data inside the file manually
- ➔ Then added the file to staging/tracking area using “git add”
- ➔ Committed the file and pushed to the repository
- ➔ Verified the file is updated and merged.

```
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ ls
1 2 Password README.md Ravutla a sample/ test.txt
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ vi 1
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add .
warning: in the working copy of '1', LF will be replaced by CRLF the next time Git touches it
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "adding file 1"
[Ravutla 0d06b3c] adding file 1
1 file changed, 1 insertion(+)
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git status
On branch Ravutla
Your branch is ahead of 'origin/Ravutla' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git push origin Ravutla
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 294 bytes | 294.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
to github.com:ravutla-10/sample.git
6ab4585..0d06b3c Ravutla -> Ravutla
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git checkout Poojith
deletion of directory 'sample' failed. Should I try again? (y/n) n
warning: unable to mkdir 'sample': Permission denied
Updating files: 100% (10/10), done.
Switched to branch 'Poojith'
Your branch is up to date with 'origin/Poojith'.
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ ls
1 2 New_Filr README.md Sample_file a file_1 file.q sample/
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ vi 1
```

```
MINGW64/d/Poojith/Devops/TEST/Sample
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git add .
warning: in the working copy of '1', LF will be replaced by CRLF the next time Git touches it
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git commit -m "adding file 1 in Poojith"
[Poojith 1104e87] adding file 1 in Poojith
1 file changed, 1 insertion(+)
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git push origin Poojith
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 299 bytes | 299.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
to github.com:ravutla-10/sample.git
13fa75..1104e87 Poojith -> Poojith
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git merge Ravutla
Auto-merging 1
CONFLICT (content): Merge conflict in 1
Automatic merge failed; fix conflicts and then commit the result.
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith/MERGING)
$ git status
On branch Poojith
Your branch is up to date with 'origin/Poojith'.

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Changes to be committed:
  new file:   Password
  new file:   Ravutla
  new file:   Sample
  new file:   test.txt

Unmerged paths:
  (use "git add <file>..." to mark resolution)
  both modified: 1
poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith/MERGING)
$ vi 1
```

MINGW64:/d/Poojith/Devops/TEST/Sample

```
<<<<<<< HEAD
hi this is file 1 in Poojith
=====
hi this is file 1 in Ravutla
>>>>>>> Ravutla
~
~
~
~
~
~
~
```

```
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith|MERGING)
$ git add .
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith|MERGING)
$ git commit -m "resolved conflict"
[Poojith 8653f07] resolved conflict
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ ls
1 2 New_fir Password README.md Ravutla Sample_file a file_1 file_q sample/ test.txt
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ cat 1
hi this is file 1 in Poojith and in Ravutla
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$
```

2. Recover Deleted Branch

- Delete a local branch and then recover it using the reflog.

- ➔ Switched to the main branch
- ➔ Checked the list of branches available using git branch
- ➔ Removed a branched “Horizon” using (**git branch -d Horizon**)
- ➔ Verified that the branch is deleted from the list
- ➔ Used **git reflog** to identify the commit related to the branch deletion
- ➔ Used (**git branch branch_name commit_ID**) to get the deleted branch back
- ➔ Verified that the branch is back and available in the list of branches (git branch)

```
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (main)
$ git branch
Horizon
Poojith
Ravutla
Techie
* main
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (main)
$ git branch -d Horizon
warning: deleting branch 'Horizon' that has been merged to
'refs/remotes/origin/Horizon', but not yet merged to HEAD.
Deleted branch Horizon (was d2e0542).
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (main)
$ git branch
Poojith
Ravutla
Techie
* main
```

```

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (main)
$ git reflog
65df115 (HEAD -> main, origin/main, origin/HEAD) HEAD@{0}: checkout: moving from Poojith to main
8653f07 (Poojith) HEAD@{1}: checkout: moving from main to Poojith
65df115 (HEAD -> main, origin/main, origin/HEAD) HEAD@{2}: checkout: moving from Poojith to main
8653f07 (Poojith) HEAD@{3}: commit (merge): resolved conflict
1104e87 (origin/Poojith) HEAD@{4}: commit: adding file 1 in Poojith
13fa775 HEAD@{5}: checkout: moving from Ravutla to Poojith
65df115 (HEAD -> main, origin/main, origin/HEAD) HEAD@{6}: commit: adding file 1
6ab4585 HEAD@{7}: commit: Test GitHub webhook
4ab9d40 HEAD@{8}: rebase (finish): returning to refs/heads/Ravutla
4ab9d40 HEAD@{9}: rebase (squash): adding ignore file
16f1c06 HEAD@{10}: rebase (start): checkout HEAD-2
865a897 HEAD@{11}: commit: adding files
16f1c06 HEAD@{12}: checkout: moving from Poojith to Ravutla
13fa775 HEAD@{13}: pull --rebase origin Poojith (finish): returning to refs/heads/Poojith
13fa775 HEAD@{14}: pull --rebase origin Poojith (pick): adding sample commit file
1359f09 HEAD@{15}: pull --rebase origin Poojith (start): checkout: 1359f097462b8d82c0762468038b59dc0418aab5
6a38c7f HEAD@{16}: checkout: moving from Poojith to Poojith
6a38c7f HEAD@{17}: rebase (finish): returning to refs/heads/Poojith
6a38c7f HEAD@{18}: rebase (squash): adding sample commit file
977bc80 HEAD@{19}: rebase (start): checkout HEAD-2
659ef52 HEAD@{20}: rebase (finish): returning to refs/heads/Poojith
659ef52 HEAD@{21}: rebase (squash): Revert "adding sample commit file"
8a740a8 HEAD@{22}: rebase (start): checkout HEAD-2
8c086ed HEAD@{23}: rebase (finish): returning to refs/heads/Poojith
8c086ed HEAD@{24}: rebase (squash): adding New_Filr
1359f09 HEAD@{25}: rebase (start): checkout HEAD-2
6a05c7d HEAD@{26}: checkout: moving from Poojith to Poojith
6a05c7d HEAD@{27}: rebase (finish): returning to refs/heads/Poojith
6a05c7d HEAD@{28}: rebase (start): checkout HEAD-2
6a05c7d HEAD@{29}: commit (amend): adding 2 new files file_1, file_q
13be307 HEAD@{30}: commit: adding 2 new files
1359f09 HEAD@{31}: commit: adding New_Filr
8a740a8 HEAD@{32}: reset: moving to HEAD
8a740a8 HEAD@{33}: reset: moving to HEAD
8a740a8 HEAD@{34}: revert: Revert "adding sample commit file"
977bc80 HEAD@{35}: commit: adding sample commit file
72ba011 HEAD@{36}: reset: moving to HEAD-3
1bc3ae5 HEAD@{37}: commit: adding reset_file to Poojith Branch
72ba011 HEAD@{38}: commit: adding ignore file
28a1a3 HEAD@{39}: checkout: moving from Ravutla to Poojith
16f1c06 HEAD@{40}: commit: adding ignore file
c7f4395 HEAD@{41}: commit: adding new file - sample
6d16a08 HEAD@{42}: checkout: moving from horizon to Ravutla
d2e0542 (origin/Horizon) HEAD@{43}: checkout: moving from Techie to Horizon
6ab876a (origin/Techie, Techie) HEAD@{44}: checkout: moving from Poojith to Techie
28a1a3 HEAD@{45}: commit: adding sample file to Poojith Branch
65df115 (HEAD -> main, origin/main, origin/HEAD) HEAD@{46}: checkout: moving from main to Poojith
65df115 (HEAD -> main, origin/main, origin/HEAD) HEAD@{47}: commit: Adding files 1 and 2
29dc173 HEAD@{48}: clone: from github.com:ravutla-10/sample.git

```

```

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (main)
$ git branch Horizon 65df115

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (main)
$ git branch
  Horizon
* Poojith
  Ravutla
  Techie
* main

```

3. Undo Wrong Push

- Push a wrong commit to GitHub, then undo it without losing history.

- ➔ Created a file commit_test
- ➔ Changed the file from untracking area to tracking area by (git add)
- ➔ Committed the file and pushed to the repository
- ➔ Identified the commit id by using **git log**
- ➔ Used git revert commit_id to undo the commit without losing the history
- ➔ Verified by using git log → previous commit and latest commit both were available.

```

MINGW64/d/Poojith/Devops/TEST/Sample

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ ls
1 2 New_Filr Password README.md Ravutla Sample_file a file_1 file_q sample/ test.txt

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ echo "this is a test file" > commit_test.txt

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ ls
1 2 New_Filr Password README.md Ravutla Sample_file a commit_test.txt file_1 file_q sample/ test.txt

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git add .
warning: in the working copy of 'commit_test.txt', LF will be replaced by CRLF the next time Git touches it

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git commit -m "adding commit_test file to Poojith branch"
[poojith eb039a4] adding commit_test file to Poojith branch
1 file changed, 1 insertion(+)
create mode 100644 commit_test.txt

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git push origin Poojith
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 309 bytes | 309.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:ravutla-10/sample.git
 8653f07..eb039a4 Poojith -> Poojith

poojith@Poojith: MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git log -3
commit eb039a46157bd7261981fb6b04dea500cd55e746 (HEAD -> Poojith, origin/Poojith)
Author: Poojith <ravutla.10@gmail.com>
Date: Fri Aug 21 14:39:05 2026 +0530

    adding commit_test file to Poojith branch

commit 8653f070defd613fe34b8fa580f528d61606cbf8
Merge: 1104e87 0d06b2c
Author: Poojith <ravutla.10@gmail.com>
Date: Fri Aug 21 12:50:23 2026 +0530

    resolved conflict

commit 1104e876a629f6d2fc2852ca4558ed3fde29600e
Author: Poojith <ravutla.10@gmail.com>
Date: Fri Aug 21 12:45:10 2026 +0530

    adding file 1 in Poojith

```

```

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git revert eb039a4e
[Pooji 6c045b7] Revert "adding commit_test file to Pooji branch"
1 file changed, 1 deletion(-)
delete mode 100644 commit_test.txt

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git push origin Pooji
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 266 bytes | 266.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
to github.com:ravutla-10/Sample.git
eb039a4..6c045b7 Pooji -> Pooji

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git log -5
commit 6c045b7a80669791087805de734148db41bb8bb0 (HEAD -> Pooji, origin/Pooji)
Author: Pooji <ravutla.10@gmail.com>
Date: Fri Aug 21 14:40:07 2026 +0530

    Revert "adding commit_test file to Pooji branch"

    This reverts commit eb039a46157bd7261961fb6b04dea500cd55e746.

commit eb039a46157bd7261961fb6b04dea500cd55e746
Author: Pooji <ravutla.10@gmail.com>
Date: Fri Aug 21 14:39:05 2026 +0530

    adding commit_test file to Pooji branch

commit 8653f070defd613fe34b8fa580f528d61606cbf8
Merge: 1104e87 0d06b3c
Author: Pooji <ravutla.10@gmail.com>
Date: Fri Aug 21 12:50:23 2026 +0530

    resolved conflict

commit 1104e876a629f6d2fc2852ca4558ed3fde29600e
Author: Pooji <ravutla.10@gmail.com>
Date: Fri Aug 21 12:45:10 2026 +0530

    adding file 1 in Pooji

commit 0d06b3c08595823e7ab2932a11b7b3e1d0231ce9 (origin/Ravutla, Ravutla)
Author: Pooji <ravutla.10@gmail.com>
Date: Fri Aug 21 12:42:37 2026 +0530

    adding file 1

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$

```

4. Amend a Commit

- Make a commit, then add a missing file to it using git commit --amend.

- ➔ Created a first file and added it to tracking area.
- ➔ Committed the first file
- ➔ Created a second file and added it to tracking area.
- ➔ Used git commit --amend --no-edit to commit the second file without changing the commit message.
- ➔ Verified that the second file is committed along with first file with same commit id.

```

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ echo "this is a amend test file 1" > amend1.txt

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git add .
warning: in the working copy of 'amend1.txt', LF will be replaced by CRLF the next time git touches it

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git status
On branch Pooji
Your branch is ahead of 'origin/Pooji' by 1 commit.
(use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   amend1.txt
    deleted:    amend1.txt
    deleted:    amend2.txt

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git commit -m "adding amend files"
[Pooji 88fe0f1] adding amend files
3 files changed, 1 insertion(+), 2 deletions(-)
create mode 100644 amend1.txt
delete mode 100644 amend1.txt
delete mode 100644 amend2.txt

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ echo "this is an amend file 2" > amend2.txt

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git add .
warning: in the working copy of 'amend2.txt', LF will be replaced by CRLF the next time git touches it

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git status
On branch Pooji
Your branch is ahead of 'origin/Pooji' by 2 commits.
(use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   amend2.txt

pooji@Pooji MINGW64 /d/Pooji/Devops/TEST/Sample (Pooji)
$ git commit --amend --no-edit
[Pooji b2ced5d] adding amend files
Date: Fri Aug 21 15:08:15 2026 +0530
4 files changed, 2 insertions(+), 2 deletions(-)
create mode 100644 amend1.txt
create mode 100644 amend2.txt
delete mode 100644 amend1.txt
delete mode 100644 amend2.txt

```

5. Cherry-pick a Commit

- Take a specific commit from one branch and apply it to another branch.

- ➔ Copied the commit id of the latest commit in Poojith Branch
- ➔ Switched to the Ravutla branch using (git checkout Ravutla)
- ➔ Used git cherry-pick commit_ID(of Poojith branch).
- ➔ Verifies that it picked the commit from previous branch

```

MINGW64:/d/Poojith/Devops/TEST/Sample

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Poojith)
$ git checkout Ravutla
Switched to branch 'Ravutla'
Your branch is up to date with 'origin/Ravutla'.

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git cherry-pick b2ced5dfe
[Ravutla 9957eb6] adding amend files
Date: Fri Aug 21 15:08:15 2026 +0530
2 files changed, 2 insertions(+)
create mode 100644 amend1.txt
create mode 100644 amend2.txt

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$

```

6. Interactive Rebase

- Reorder and squash multiple commits into a single clean commit.

```

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ echo "this is rebase 1" > rebase1.txt

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add .
warning: in the working copy of 'rebase1.txt', LF will be replaced by CRLF the next time Git touches it
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "adding rebase1"
[Ravutla 23d80d3] adding rebase1
1 file changed, 1 insertion(+)
create mode 100644 rebase1.txt

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ echo "this is rebase 2" > rebase2.txt

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add .
warning: in the working copy of 'rebase2.txt', LF will be replaced by CRLF the next time Git touches it
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "adding rebase 2"
[Ravutla 23d80d3] adding rebase 2
1 file changed, 1 insertion(+)
create mode 100644 rebase2.txt

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ echo "this is rebase 3" > rebase3.txt

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add .
warning: in the working copy of 'rebase3.txt', LF will be replaced by CRLF the next time Git touches it
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "adding rebase3"
[Ravutla 0d40848] adding rebase3
1 file changed, 1 insertion(+)
create mode 100644 rebase3.txt

```

```

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log -4
commit 0d408482e267340774c20875e6a6120a23da
Author: pooji@ravutla.100gm1.com
Date: Fri Aug 21 15:08:40 2026 +0530
    adding rebase3

commit 23d80d3280322d8031728915a7f026420604
Author: pooji@ravutla.100gm1.com
Date: Fri Aug 21 15:11:04 2026 +0530
    adding rebase 2

commit 23d80d3280322d8031728915a7f026420604
Author: pooji@ravutla.100gm1.com
Date: Fri Aug 21 15:20:40 2026 +0530
    adding rebase1

commit 9957eb63af7e64071567f9405120a23da
Author: pooji@ravutla.100gm1.com
Date: Fri Aug 21 15:08:15 2026 +0530
    adding amend files

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git rebase -i HEAD~4
[Rebasing: 4/4] adding all 3 rebase files
Date: Fri Aug 21 15:20:40 2026 +0530
3 files changed, 3 insertions(+)
create mode 100644 rebase1.txt
create mode 100644 rebase2.txt
create mode 100644 rebase3.txt
Successfully rebased and updated refs/heads/Ravutla.

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log -4
commit 0d408482e267340774c20875e6a6120a23da
Author: pooji@ravutla.100gm1.com
Date: Fri Aug 21 15:08:40 2026 +0530
    adding all 3 rebase files

commit 9957eb63af7e64071567f9405120a23da
Author: pooji@ravutla.100gm1.com
Date: Fri Aug 21 15:20:40 2026 +0530
    adding amend files

commit 0d408482e267340774c20875e6a6120a23da (origin/Ravutla)
Author: pooji@ravutla.100gm1.com
Date: Fri Aug 21 15:08:15 2026 +0530
    adding file 1

```

```
MINGW64/d/Poojith/Devops/TEST/Sample
pick adaa6ba adding rebase1
pick 23d80e3 adding rebase 2
pick 0d40d48 adding rebase1
# Rebase 9957eb6..0d40d48 onto 9957eb6 (3 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-C] [-c] <commit> = like "squash" but keep only the previous
#   commit's log message, unless -C is used, in which case
#   keep only this commit's message; -c is same as -C but
#   opens the editor
# x, exec <command> = run command (the rest of the line) using shell
# b, break = stop here (continue rebase later with 'git rebase --continue')
# d, drop <commit> = remove commit
# l, label <label> = label current HEAD with a name
# t, reset <label> = reset HEAD to a label
# m, merge [-C] [-c] <commit> [-C] <label> [# <oneline>]
#   create a merge commit using the original merge commit's
#   message (or the oneline, if no original merge commit was
#   specified); use -C <commit> to reword the commit message
#
# These lines can be re-ordered; they are executed from top to bottom.
#
# If you remove a line here THAT COMMIT WILL BE LOST.
#
# However, if you remove everything, the rebase will be aborted.
#
```

7. Tagging & Release

- Create a version tag (v1.0), push it to GitHub, then delete and restore it.

- ➔ Created a version tag (v1.0) – git tag v1.0
- ➔ Pushed the tag to the remote (GitHub)
- ➔ Deleted the version tag from local and remote
- ➔ Restored the version tag using the latest commit id
- ➔ Verified that the version is restored and pushed it into GitHub.

```
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log --oneline
e5bbea4 (HEAD -> Ravutla) adding all 3 rebase files
9957eb6 adding amend files
0d06b3c (origin/Ravutla) adding file 1
6ab4585 Test Github webhook
4ab9df0 adding ignore file
e2fa395 adding new file - sample
6416aa8 Create Ravutla
65df115 (origin/main, origin/HEAD, main, Horizon) Adding files 1 and 2
230c173 Merge branch 'main' of github.com:ravutla-10/Sample
3f6f81c Added ignore file
bf9326a Add initial README with project description
24b697c Adding a

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git tag v1.0

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git tag
v1.0

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git push origin v1.0
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (9/9), 714 bytes | 714.00 KiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:ravutla-10/Sample.git
 * [new tag]         v1.0 -> v1.0

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git tag -d v1.0
Deleted tag 'v1.0' (was e5bbea4)

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git push origin --delete v1.0
To github.com:ravutla-10/Sample.git
- [deleted]         v1.0
```

```
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log --oneline
e5bbea4 (HEAD -> Ravutla) adding all 3 rebase files
9957eb6 adding amend files
0d06b3c (origin/Ravutla) adding file 1
6ab4585 Test Github webhook
4ab9df0 adding ignore file
e2fa395 adding new file - sample
6416aa8 Create Ravutla
65df115 (origin/main, origin/HEAD, main, Horizon) Adding files 1 and 2
230c173 Merge branch 'main' of github.com:ravutla-10/Sample
3f6f81c Added ignore file
bf9326a Add initial README with project description
24b697c Adding a

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git tag v1.0 e5bbea4

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git push origin v1.0
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (9/9), 714 bytes | 714.00 KiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:ravutla-10/Sample.git
 * [new tag]         v1.0 -> v1.0

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git show v1.0
commit e5bbea435d927c83f33c675532cb64bd1f42217f (HEAD -> Ravutla, tag: v1.0)
Author: Poojith <ravutla.10@gmail.com>
Date: Fri Aug 21 15:20:40 2026 +0530

    adding all 3 rebase files
```

8. Clone with Sparse Checkout

- Clone only a subdirectory of a repo using sparse checkout.
- ➔ Clone the repository without checking out files (`git clone --no-checkout git@github.com:ravutla-10/Sample.git SparseSample`)
- ➔ Entered inside SparseSample directory
- ➔ Switched the branch using `git checkout`
- ➔ Used **git sparse-checkout init** to initialize
- ➔ `git sparse-checkout set --no-cone 'sub_DIR/'` → for sub directory
- ➔ verified that only one sub directory is visible inside SparseSample

```
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST
$ pwd
/d/Poojith/Devops/TEST

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST
$ git clone --no-checkout git@github.com:ravutla-10/Sample.git SparseSample
Cloning into 'SparseSample'...
remote: Enumerating objects: 69, done.
remote: Counting objects: 100% (69/69), done.
remote: Compressing objects: 100% (43/43), done.
remote: Total 69 (delta 18), reused 48 (delta 8), pack-reused 0 (from 0)
Receiving objects: 100% (69/69), 9.87 KiB | 1.97 MiB/s, done.
Resolving deltas: 100% (18/18), done.

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST
$ cd SparseSample/

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/SparseSample (main)
$ git branch -a
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/Horizon
  remotes/origin/Poojith
  remotes/origin/Ravutla
  remotes/origin/Techie
  remotes/origin/clone_branch
  remotes/origin/main

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/SparseSample (main)
$ git checkout Ravutla
Switched to a new branch 'Ravutla'
branch 'Ravutla' set up to track 'origin/Ravutla'.

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/SparseSample (Ravutla)
$ git sparse-checkout init
```

```
pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/SparseSample (Ravutla|SPARSE)
$ git sparse-checkout set --no-cone 'sub_DIR/'

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/SparseSample (Ravutla|SPARSE)
$ git sparse-checkout list
sub_DIR/

pooji@Poojith MINGW64 /d/Poojith/Devops/TEST/SparseSample (Ravutla|SPARSE)
$ ls
sub_DIR/
```

9. Reset vs Revert Challenge

- Demonstrate the difference between `git reset --hard` and `git revert` in a repo.
- ➔ `Git reset --hard` is used when you want to reset the commit without saving changes
 - `git reset --hard HEAD~1` (to reset the latest commit)
 - It will remove the latest commit.
- ➔ `Git revert` is used to revert the committed command and also save changes.
 - This is used when you want to make changes of any commit and also save the changes.

```

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add a
poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git status
On branch Ravutla
Your branch is ahead of 'origin/Ravutla' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   a

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        b

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "adding file a"
[ Ravutla 1b80d86] adding file a
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 a

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log -3
commit 1b80d860604b39137b8623ed23e9e66c80a0baf6 (HEAD -> Ravutla)
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 17:13:58 2026 +0530

    adding file a

commit 804b7df1d5083e03c2bba47245a865c5c19acb4
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 17:13:08 2026 +0530

    deleted files

commit cd4e93b123db2f8cc3077b3b67911a3c6a2d018c (origin/Ravutla)
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 16:09:07 2026 +0530

    adding new directory and files

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git reset --hard HEAD~1
HEAD is now at 804b7df deleted files

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log -2
commit 804b7df1d5083e03c2bba47245a865c5c19acb4 (HEAD -> Ravutla)
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 17:13:08 2026 +0530

    deleted files

commit cd4e93b123db2f8cc3077b3b67911a3c6a2d018c (origin/Ravutla)

```

```

commit cd4e93b123db2f8cc3077b3b67911a3c6a2d018c (origin/Ravutla)
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 16:09:07 2026 +0530

    adding new directory and files

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git status
On branch Ravutla
Your branch is ahead of 'origin/Ravutla' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        b

nothing added to commit but untracked files present (use "git add" to track)

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add .
poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "adding file b"
[ Ravutla b8b6e5f] adding file b
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 b

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git revert b8b6e5f
[ Ravutla 42ec173] Revert "adding file b"
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 b

```

```

MINGW64/d/Poojith/Devops/TEST/Sample
Revert "adding file b"

This reverts commit b8b6e5f6079ea55908d2aec4da6dcaab428e0421.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch Ravutla
# Your branch is ahead of 'origin/Ravutla' by 2 commits.
#   (use "git push" to publish your local commits)
#
# Changes to be committed:
#       deleted:   b
#
~
~
~
~
~
~
~
~

```

```

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log -3
commit 42ec17329b9a06b21765473d1cd07328f6bb3821 (HEAD -> Ravutla)
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 17:20:12 2026 +0530

    Revert "adding file b"

    This reverts commit b8b6e5f6079ea55908d2aec4da6dcaab428e0421.

commit b8b6e5f6079ea55908d2aec4da6dcaab428e0421
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 17:19:47 2026 +0530

    adding file b

commit 804b7df1d5083e03c2bba47245a865c5c19acb4
Author: Poojith <ravutla.10@gmail.com>
Date:   Fri Aug 21 17:13:08 2026 +0530

    deleted files

poojith@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)

```


10. Detached HEAD Challenge

- Checkout a specific commit (detached HEAD state) and create a new branch from it.

- ➔ Checked the commit history using `git log --oneline`
- ➔ Checked out a specific commit using `git checkout`.
- ➔ Verified the detached HEAD state using `git status`.
- ➔ Created a new branch from the current commit.
- ➔ Switched to the newly created branch.
- ➔ Verified the branch.

```
poojith@poojith:~/DEVOPS/TEST/Sample (Ravutla)
$ git log --oneline
217b0ae (HEAD -> Ravutla) adding file a
42ec173 Revert "adding file b"
b8b6e5f adding file b
804b7df deleted files
e4e92b (origin/Ravutla) adding new directory and files
e5bbea4 (tag: v1.0) adding all 3 rebase files
9957eb6 adding amend files
606b02c adding file 1
6ab4585 Test Github webhook
4ab9df0 adding ignore file
e7a3d95 adding new file - sample
6416aa8 Create Ravutla
65df115 (origin/main, origin/HEAD, main, Horizon) Adding files 1 and 2
22c173 Merge branch 'main' of github.com:ravutla-10/Sample
376f81c Added ignore file
bf9326a Add initial README with project description
24b697c Adding a

poojith@poojith:~/DEVOPS/TEST/Sample (Ravutla)
$ git checkout ^C

poojith@poojith:~/DEVOPS/TEST/Sample (Ravutla)
$ git checkout 804b7df
Note: switching to '804b7df'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false
HEAD is now at 804b7df deleted files

poojith@poojith:~/DEVOPS/TEST/Sample ((804b7df...))
$ git status
HEAD detached at 804b7df
nothing to commit, working tree clean

poojith@poojith:~/DEVOPS/TEST/Sample ((804b7df...))
$ git switch -c recovery_branch
Switched to a new branch 'recovery_branch'
```

```
poojith@poojith:~/DEVOPS/TEST/Sample ((804b7df...))
$ git switch -c recovery_branch
Switched to a new branch 'recovery_branch'

poojith@poojith:~/DEVOPS/TEST/Sample (recovery_branch)
$ git branch
  Horizon
  poojith
  Ravutla
  Techie
* main
+ recovery_branch

poojith@poojith:~/DEVOPS/TEST/Sample (recovery_branch)
$
```

11. Git Hooks Challenge

- Configure a pre-commit hook to reject commits without a message format (e.g., must start with JIRA-XXX).

- ➔ Created a commit-msg hook inside `.git/hooks`.
- ➔ Added validation to check that the commit message starts with JIRA-XXX.
- ➔ Made the hook executable using `chmod +x`.
- ➔ Attempted a commit with an invalid message.
- ➔ Verified that the commit was rejected.
- ➔ Created a valid commit such as JIRA-1 testing git hooks.
- ➔ Verified that the valid commit was accepted.

```

root@Poojith: /d/Poojith/Devops/TEST/Sample/.git/hooks (GIT_DIR)
$ nano .git/hooks/commit-msg
root@Poojith: /d/Poojith/Devops/TEST/Sample/.git/hooks (GIT_DIR)
$ ls
applypatch-msg.sample*  commit-msg.sample*  post-update.sample*  pre-commit.sample*  pre-push.sample*  pre-receive.sample*  push-to-checkout.sample*
commit-msg*             fsmonitor-watchman.sample*  pre-applypatch.sample*  pre-merge-commit.sample*  pre-rebase.sample*  prepare-commit-msg.sample*  update.sample*
root@Poojith: /d/Poojith/Devops/TEST/Sample/.git/hooks (GIT_DIR)
$ chmod 755 commit-msg
root@Poojith: /d/Poojith/Devops/TEST/Sample/.git/hooks (GIT_DIR)
$ cd ../..
root@Poojith: /d/Poojith/Devops/TEST/Sample (Ravutla)
$ echo "test" > hook.txt
root@Poojith: /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add .
warning: in the working copy of 'hook.txt', LF will be replaced by CRLF the next time Git touches it
root@Poojith: /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "testing git hooks"
ERROR: Commit message must start with JIRA-XXX
root@Poojith: /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "JIRA-1 testing git hooks"
(Ravutla 32edeb7) JIRA-1 testing git hooks
1 file changed, 1 insertion(+)
create mode 100644 hook.txt

```

12. Squash Merge vs Rebase Merge

- Show the difference between squash merge and rebase merge with evidence.

- ➔ Created a feature branch.
- ➔ Created multiple commits on the feature branch.
- ➔ Pushed the branch to GitHub.
- ➔ Created a Pull Request.
- ➔ Used Squash and merge and checked the resulting history.
- ➔ Created another feature branch with multiple commits.
- ➔ Used Rebase and merge.
- ➔ Compared both merge results.

```

root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (main)
$ git pull origin main
From github.com:erjonhashani-10/contribute-practice
* branch      main      -> FETCH_HEAD
Already up to date.
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (main)
$ git switch -c feature/squash-demo
Switched to a new branch 'feature/squash-demo'
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ git branch
* feature/squash-demo
  main
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ nano README.md
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ git status
On branch feature/squash-demo
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ git add README.md
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ git commit -m "Adding squash demo section"
[feature/squash-demo 4a071f6] Adding squash demo section
1 file changed, 4 insertions(+)
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ nano README.md
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ git add README.md
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ git commit -m "Adding second squash demo change"
[feature/squash-demo e7b8be7] Adding second squash demo change
1 file changed, 3 insertions(+)
root@Poojith: /d/Poojith/Devops/TEST/contribute-practice (feature/squash-demo)
$ git log --oneline --graph --decorate -5
* 4a071f6 Adding squash demo section
* e95d89b (origin/main, origin/HEAD, main) Merge pull request #58 from ErjonHashani/contribute-practice
* e8e73bd Update README.md
* d9d6664 Update contributing.md

```

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base: main ← compare: feature/squash-demo ✓ Able to merge. These branches can be automatically merged.

Discuss and review the changes in this comparison with others. [Learn about pull requests](#) [Create pull request](#)

2 commits 1 file changed 1 contributor

Commits on Aug 24, 2026

Adding squash demo section
ravutla-10 committed 5 minutes ago 4a071f6

Adding second squash demo change
ravutla-10 committed 5 minutes ago e7b8be7

Showing 1 changed file with 7 additions and 0 deletions. [Split](#) [Unified](#)

7 README.md

@@ -364,3 +364,10 @@ denisfrd (https://github.com/denisfrd)

364 364 <!-- @cassiofb-dev: Thanks for the initiative HyunCafe it was cool and fun! -->

365 365 <!-- @cassiofb-dev: I think it would be funny to keep this comments -->

366 366 <!-- @cassiofb-dev: People will only see this with the source code :p -->

367 +

368 + ## Squash Merge Demo

369 +

Squash Merge Demo #1

Merged ravutla-10 merged 2 commits into main from feature/squash-demo 1 minute ago

Code

Conversation 0 Commits 2 Checks 0 Files changed 1 +7

Owner

ravutla-10 commented 2 minutes ago

This pull request is created to demonstrate squash merge.
It contains two separate commits that will be combined into one commit using Squash and merge.

Added 2 commits 10 minutes ago

Adding squash demo section 4a071f6

Adding second squash demo change e7b8be7

ravutla-10 merged commit 39edd3a into main 1 minute ago

Revert

Reviews
No reviews
Still in progress? [Convert to draft](#)

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

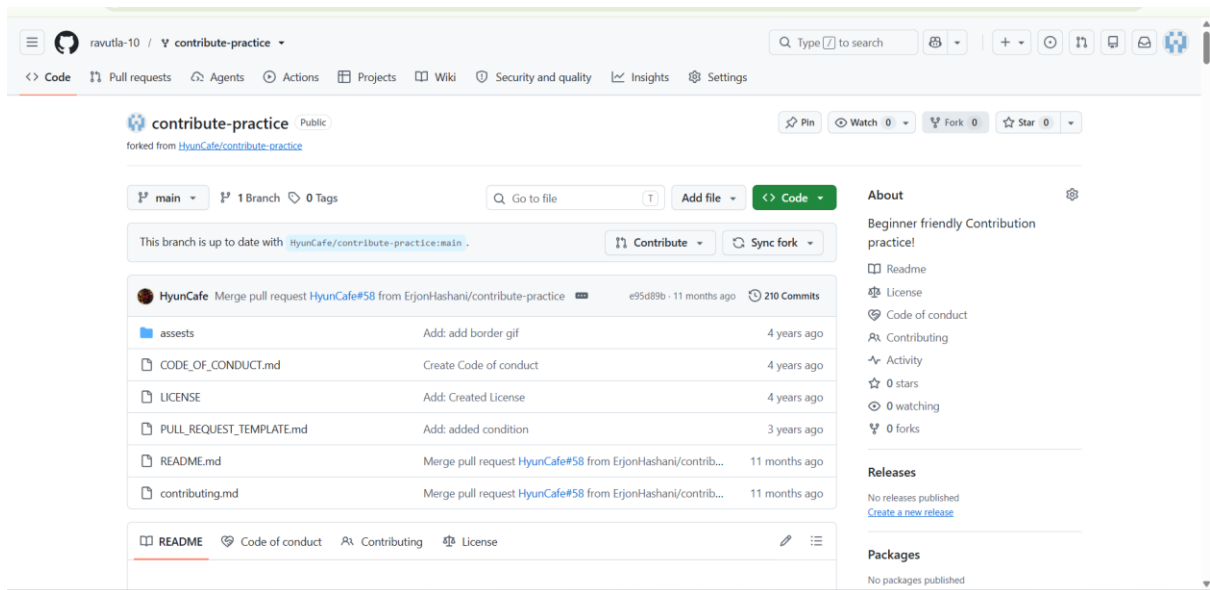
Development
Successfully merging this pull request may close these issues.
None yet

Pull request successfully merged and closed
You're all set — the feature/squash-demo branch can be safely deleted.

Delete branch

13. Fork & Pull Request Workflow

- Fork a repo, make a change, and submit a pull request to the original repo.
- ➔ Selected an open-source GitHub repository.
- ➔ Forked the repository to the GitHub account.
- ➔ Cloned the forked repository locally.
- ➔ Created a new feature branch.
- ➔ Made the required changes.
- ➔ Added and committed the changes.
- ➔ Pushed the feature branch to the forked repository.
- ➔ Created a Pull Request from the fork to the original repository.
- ➔ Reviewed the changes and submitted the Pull Request.



```
poojith@Poojith: ~/d/Poojith/Devops/TEST
$ git clone git@github.com:ravutla-10/contribute-practice.git
Cloning into 'contribute-practice'...
remote: Enumerating objects: 577, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (93/93), done.
remote: Total 577 (delta 64), reused 39 (delta 31), pack-reused 453 (from 2)
Receiving objects: 100% (577/577), 16.45 MiB | 2.86 MiB/s, done.
Resolving deltas: 100% (311/311), done.

poojith@Poojith: ~/d/Poojith/Devops/TEST
$ cd contribute-practice/

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (main)
$ git remote -v
origin git@github.com:ravutla-10/contribute-practice.git (fetch)
origin git@github.com:ravutla-10/contribute-practice.git (push)

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (main)
$
```

```
poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (main)
$ git switch -c feature/devops-update
Switched to a new branch 'feature/devops-update'

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git branch
* feature/devops-update
main

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ ls
CODE_OF_CONDUCT.md LICENSE PULL_REQUEST_TEMPLATE.md README.md assests/ contributing.md

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ nano README.md

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git status
On branch feature/devops-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git diff
diff --git a/README.md b/README.md
index b53e824..91407e4 100644
--- a/README.md
+++ b/README.md
@@ -32,6 +32,9 @@ I have prepared a set of straightforward tasks to guide you through each step of

## Clone this repo

## DevOps Practice

<This change was made to practice the GitHub Fork and Pull Request workflow.
 <br>

<h2>Step 1</h2>

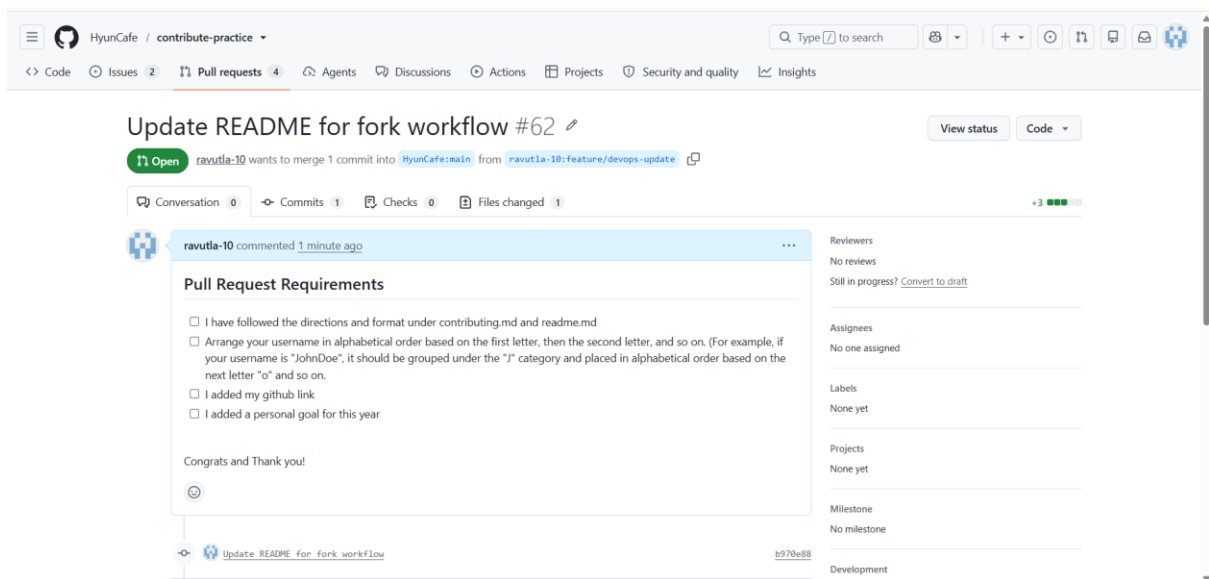
poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git add README.md

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git status
On branch feature/devops-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md
```

```
poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git commit -m "Update README for fork workflow"
[feature/devops-update b570e88] Update README for fork workflow
1 file changed, 3 insertions(+)

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git status
On branch feature/devops-update
nothing to commit, working tree clean

poojith@Poojith: ~/d/Poojith/Devops/TEST/contribute-practice (feature/devops-update)
$ git push -u origin feature/devops-update
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 399 bytes | 399.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'feature/devops-update' on GitHub by visiting:
remote:   https://github.com/ravutla-10/contribute-practice/pull/new/feature/devops-update
remote:
To github.com:ravutla-10/contribute-practice.git
 * [new branch]   feature/devops-update -> feature/devops-update
branch 'feature/devops-update' set up to track 'origin/feature/devops-update'.
```



14. Recover Lost Commit

- Commit something, reset hard, and then recover it using git reflog.

- ➔ Created a file and committed it.
- ➔ Verified the commit using git log.
- ➔ Used git reset --hard HEAD~1.
- ➔ Verified that the commit was no longer visible in the normal Git history.
- ➔ Used git reflog to find the previous commit.
- ➔ Identified the lost commit ID.
- ➔ Created a new branch from the recovered commit.
- ➔ Verified the recovered file and commit history.

```

poorj@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ echo "Important work" > important.txt
poorj@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git add important.txt
warning: in the working copy of 'important.txt', LF will be replaced by CRLF the next time Git touches it
poorj@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git commit -m "JIRA-123 Important work"
[Ravutla 4427c49] JIRA-123 Important work
1 file changed, 1 insertion(+)
create mode 100644 important.txt
poorj@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log --oneline
4427c49 (HEAD -> Ravutla) JIRA-123 Important work
32edcbf JIRA-1 testing git hooks
21fb0ae (origin/Ravutla) adding file a
42ac173 Revert "adding file b"
b8b6e5f adding file b
804b7df (recovery_branch) deleted files
c9e93b adding new directory and files
e5bbea4 (tag: v1.0) adding all 3 rebase files
9957eb6 adding amend files
60b0b3c adding file 1
6ab4585 Test Github webhook
4ab9d70 adding ignore file
c7a3b95 adding new file - sample
6416aa8 Create Ravutla
65df115 (origin/main, origin/HEAD, main, Horizon) Adding files 1 and 2
32ac173 Merge branch 'main' of github.com:ravutla-10/sample
3f6f81c Added ignore file
b9326e Add initial README with project description
24b697c Adding a
poorj@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git reset --hard HEAD~1
HEAD is now at 32edcbf JIRA-1 testing git hooks
poorj@Poojith MINGW64 /d/Poojith/Devops/TEST/Sample (Ravutla)
$ git log --oneline
32ac173 (HEAD -> Ravutla) JIRA-1 testing git hooks
21fb0ae (origin/Ravutla) adding file a
42ac173 Revert "adding file b"
b8b6e5f adding file b
804b7df (recovery_branch) deleted files
c9e93b adding new directory and files
e5bbea4 (tag: v1.0) adding all 3 rebase files
9957eb6 adding amend files
60b0b3c adding file 1
6ab4585 Test Github webhook
4ab9d70 adding ignore file
c7a3b95 adding new file - sample
6416aa8 Create Ravutla
65df115 (origin/main, origin/HEAD, main, Horizon) Adding files 1 and 2
32ac173 Merge branch 'main' of github.com:ravutla-10/sample
3f6f81c Added ignore file

```

```
poojith@poojith:~/devops/TEST/sample$ git reflog
32edc49 HEAD -> Ravutla HEAD@0: reset: moving to HEAD-1
4427c49 HEAD@1: commit: JIRA-123 Important work
32edc49 HEAD -> Ravutla HEAD@2: commit: JIRA-1 testing git hooks
21fb0ae (origin/Ravutla) HEAD@3: checkout: moving from recovery_branch to Ravutla
804b7df (recovery_branch) HEAD@4: checkout: moving from Ravutla to recovery_branch
21fb0ae (origin/Ravutla) HEAD@5: checkout: moving from recovery_branch to Ravutla
804b7df (recovery_branch) HEAD@6: checkout: moving from Ravutla to recovery_branch
804b7df (recovery_branch) HEAD@7: checkout: moving from Ravutla to 804b7df
21fb0ae (origin/Ravutla) HEAD@8: reset: moving to 21fb0ae
f12a41 HEAD@9: revert: Revert "adding file b"
42ec173 HEAD@10: reset: moving to HEAD-1
21fb0ae (origin/Ravutla) HEAD@11: commit: adding file a
42ec173 HEAD@12: revert: Revert "adding file b"
b8b6e5f HEAD@13: commit: adding file b
804b7df (recovery_branch) HEAD@14: reset: moving to HEAD-1
18a0d86 HEAD@15: commit: adding file a
804b7df (recovery_branch) HEAD@16: commit: deleted files
c4de93b HEAD@17: commit: adding new directory and files
e5bbea4 (tag: v1.0) HEAD@18: rebase (finish): returning to refs/heads/Ravutla
e5bbea4 (tag: v1.0) HEAD@19: rebase (squash): adding all 3 rebase files
a6cc2c6 HEAD@20: rebase (squash): # This is a combination of 2 commits.
adaa66a HEAD@21: rebase (start): checkout HEAD-3
0d40d48 HEAD@22: commit: adding rebase3
23d80d3 HEAD@23: commit: adding rebase 2
adaa66a HEAD@24: commit: adding rebase1
995eb66 HEAD@25: cherry-pick: adding amend files
0d06b3c HEAD@26: checkout: moving from Poojith to Ravutla
32edc49 (Poojith) HEAD@27: commit (amend): adding amend files
88fe0f1 HEAD@28: commit: adding amend files
882c302 HEAD@29: commit (amend): adding two amend files
427bfde HEAD@30: commit (amend): adding amend file 1
5be648a HEAD@31: commit (amend): adding amend file 1
01f6a86 HEAD@32: commit: adding amend file 1
e0491b7 (origin/Poojith) HEAD@33: revert: Revert "adding commit test file to Poojith branch"
eb039a4 HEAD@34: commit: adding commit test file to Poojith branch
8653f07 HEAD@35: checkout: moving from Ravutla to Poojith
a080d3c HEAD@36: checkout: moving from main to Ravutla
03df115 (origin/main, origin/HEAD, main, horizon) HEAD@37: checkout: moving from Poojith to main
8653f07 HEAD@38: checkout: moving from main to Poojith
03df115 (origin/main, origin/HEAD, main, horizon) HEAD@39: checkout: moving from Poojith to main
8653f07 HEAD@40: commit (merge): resolved conflict
1104e87 HEAD@41: commit: adding file 1 in Poojith
13fa775 HEAD@42: checkout: moving from Ravutla to Poojith
0a90b3c HEAD@43: commit: adding file 1
0ba4585 HEAD@44: commit: Test Github webhook
```

```
poojith@poojith:~/devops/TEST/sample$ git reset --hard 4427c49
HEAD is now at 4427c49 JIRA-123 Important work

poojith@poojith:~/devops/TEST/sample$ git reflog
4427c49 HEAD -> Ravutla HEAD@0: reset: moving to 4427c49
32edc49 HEAD@1: reset: moving to HEAD-1
4427c49 HEAD -> Ravutla HEAD@2: commit: JIRA-123 Important work
32edc49 HEAD@3: commit: JIRA-1 testing git hooks
21fb0ae (origin/Ravutla) HEAD@4: checkout: moving from recovery_branch to Ravutla
804b7df (recovery_branch) HEAD@5: checkout: moving from Ravutla to recovery_branch
21fb0ae (origin/Ravutla) HEAD@6: checkout: moving from recovery_branch to Ravutla
804b7df (recovery_branch) HEAD@7: checkout: moving from Ravutla to 804b7df
804b7df (recovery_branch) HEAD@8: checkout: moving from Ravutla to 804b7df
21fb0ae (origin/Ravutla) HEAD@9: reset: moving to 21fb0ae
f12a41 HEAD@10: revert: Revert "adding file b"
42ec173 HEAD@11: reset: moving to HEAD-1
21fb0ae (origin/Ravutla) HEAD@12: commit: adding file a
42ec173 HEAD@13: revert: Revert "adding file b"
b8b6e5f HEAD@14: commit: adding file b
804b7df (recovery_branch) HEAD@15: reset: moving to HEAD-1
18a0d86 HEAD@16: commit: adding file a
804b7df (recovery_branch) HEAD@17: commit: deleted files
c4de93b HEAD@18: commit: adding new directory and files
e5bbea4 (tag: v1.0) HEAD@19: rebase (finish): returning to refs/heads/Ravutla
e5bbea4 (tag: v1.0) HEAD@20: rebase (squash): adding all 3 rebase files
a6cc2c6 HEAD@21: rebase (squash): # This is a combination of 2 commits.
adaa66a HEAD@22: rebase (start): checkout HEAD-3
0d40d48 HEAD@23: commit: adding rebase3
23d80d3 HEAD@24: commit: adding rebase 2
adaa66a HEAD@25: commit: adding rebase1
995eb66 HEAD@26: cherry-pick: adding amend files
0d06b3c HEAD@27: checkout: moving from Poojith to Ravutla
32edc49 (Poojith) HEAD@28: commit (amend): adding amend files
88fe0f1 HEAD@29: commit: adding amend files
882c302 HEAD@30: commit (amend): adding two amend files
427bfde HEAD@31: commit (amend): adding amend file 1
5be648a HEAD@32: commit (amend): adding amend file 1
01f6a86 HEAD@33: commit: adding amend file 1
e0491b7 (origin/Poojith) HEAD@34: revert: Revert "adding commit test file to Poojith branch"
eb039a4 HEAD@35: commit: adding commit test file to Poojith branch
8653f07 HEAD@36: checkout: moving from Ravutla to Poojith
a080d3c HEAD@37: checkout: moving from main to Ravutla
03df115 (origin/main, origin/HEAD, main, horizon) HEAD@38: checkout: moving from Poojith to main
```